Transformations & Symmetry

ClipSAT Printable Practice Suite (clipsat.org) • Topic 9.01

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 [Foundational] • Target: Properties of rectangles, rhombuses, and squares

A geometric figure scaled by factor $k = 2$ has an original side length of 8 cm. What is the corresponding image length? Show your complete algebraic work.

2 [Foundational] • Target: Properties of rectangles, rhombuses, and squares

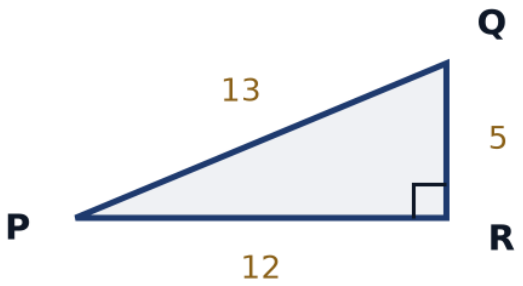
On the coordinate plane, a line with slope $m = 4$ passes through $(0, 8)$. What is its y-coordinate when $x = 1$? Show your complete algebraic work.

3 [Foundational] • Target: Properties of rectangles, rhombuses, and squares

In ΔABC studying properties of rectangles, rhombuses, and squares, $m\angle A = 34^\circ$ and $m\angle B = 52^\circ$. What is $m\angle C$? Show your complete algebraic work.

4 [Foundational] • Target: Properties of rectangles, rhombuses, and squares

A geometric figure scaled by factor $k = 3$ has an original side length of 8 cm. What is the corresponding image length? Show your complete algebraic work.



5 [Foundational] • Target: Properties of rectangles, rhombuses, and squares

On the coordinate plane, a line with slope $m = 4$ passes through $(0, 8)$. What is its y-coordinate when $x = 2$? Show your complete algebraic work.

6 [Foundational] • Target: Properties of rectangles, rhombuses, and squares

In ΔABC studying properties of rectangles, rhombuses, and squares, $m\angle A = 34^\circ$ and $m\angle B = 59^\circ$. What is $m\angle C$? Show your complete algebraic work.

7 [Foundational] • Target: Properties of rectangles, rhombuses, and squares

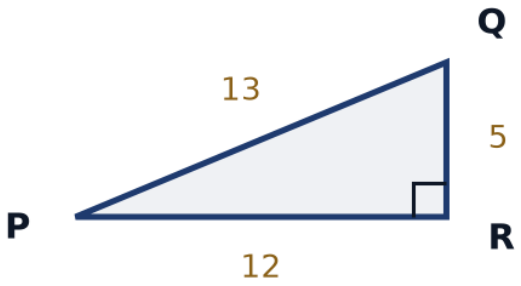
A geometric figure scaled by factor $k = 4$ has an original side length of 8 cm. What is the corresponding image length? Show your complete algebraic work.

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Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

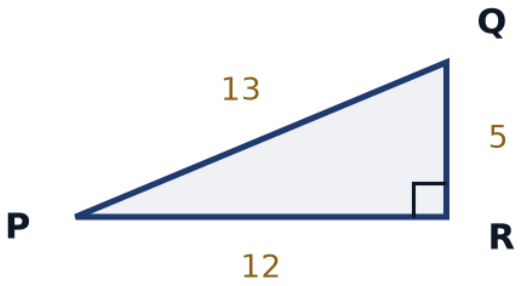
8 [Intermediate] • Target: Properties of rectangles, rhombuses, and squares

On the coordinate plane, a line with slope $m = 4$ passes through $(0, 8)$. What is its y-coordinate when $x = 3$? Show your complete algebraic work.



9 [Intermediate] • Target: Properties of rectangles, rhombuses, and squares

In ΔABC studying properties of rectangles, rhombuses, and squares, $m\angle A = 34^\circ$ and $m\angle B = 66^\circ$. What is $m\angle C$? Show your complete algebraic work.



10 [Intermediate] • Target: Properties of rectangles, rhombuses, and squares

A geometric figure scaled by factor $k = 5$ has an original side length of 8 cm. What is the corresponding image length? Show your complete algebraic work.

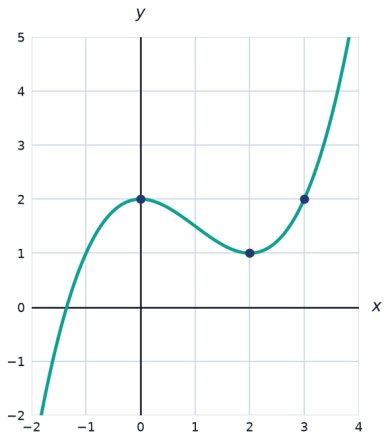
11 [Intermediate] • Target: Properties of rectangles, rhombuses, and squares

On the coordinate plane, a line with slope $m = 4$ passes through $(0, 8)$. What is its y-coordinate when $x = 4$? Show your complete algebraic work.

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12 [Intermediate] • Target: Properties of trapezoids

For the model $f(x) = 2x + 9$ in properties of trapezoids, evaluate $f(2)$.



13 [Intermediate] • Target: Properties of trapezoids

Find the positive solutions to the quadratic equation $x^2 - 7x + 6 = 0$.

14 [Intermediate] • Target: Properties of trapezoids

A data set recorded for properties of trapezoids consists of the four values $\{11, 15, 19, 25\}$. What is the arithmetic mean? Show your complete algebraic work.

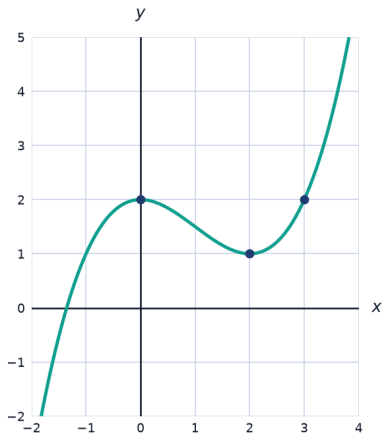
Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 [Challenging] • Target: Properties of trapezoids • Cross-Lesson: 3.01 Triangles & Congruence

For the model $f(x) = 2x + 9$ in properties of trapezoids, evaluate $f(3)$.

16 [Challenging] • Target: Properties of trapezoids • Cross-Lesson: 4.01 Similarity & Proportional Reasoning

Find the positive solutions to the quadratic equation $x^2 - 9x + 14 = 0$.



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17 [Challenging] • Target: Properties of trapezoids • Cross-Lesson: 5.01 Right Triangles & Trigonometry

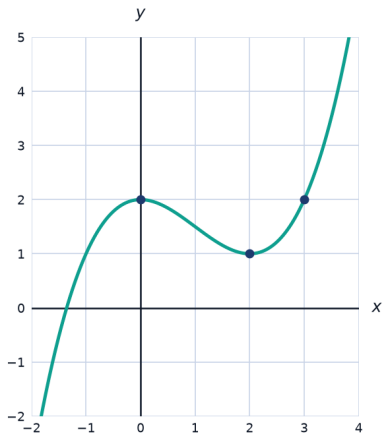
For the model $f(x) = 2x + 9$ in properties of trapezoids, evaluate $f(4)$.

18 [Challenging] • Target: Properties of trapezoids • Cross-Lesson: 6.01 Polygons & Quadrilaterals

Find the positive solutions to the quadratic equation $x^2 - 11x + 24 = 0$.

19 [Challenging] • Target: Properties of trapezoids • Cross-Lesson: 7.01 Circles

For the model $f(x) = 2x + 9$ in properties of trapezoids, evaluate $f(5)$.



20 [Challenging] • Target: Properties of trapezoids • Cross-Lesson: 8.01 Area, Surface Area & Volume

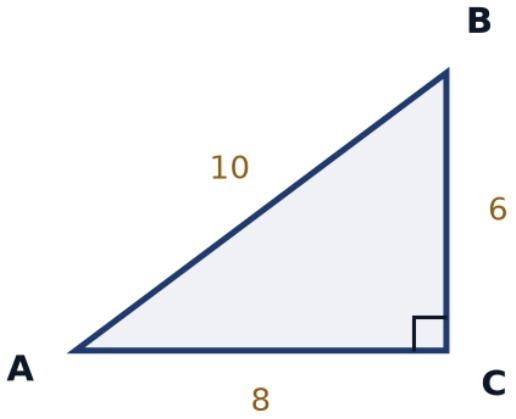
Find the positive solutions to the quadratic equation $x^2 - 13x + 36 = 0$.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

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21 [Foundational] • Target: Angles of polygons

In ΔABC studying angles of polygons, $m\angle A = 32^\circ$ and $m\angle B = 45^\circ$. What is $m\angle C$?



a. 103°

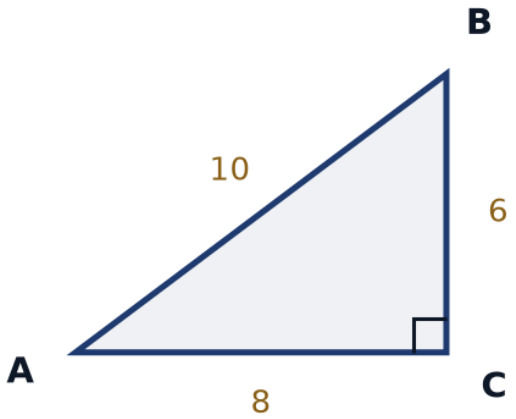
b. 113°

c. 95°

d. 148°

22 [Foundational] • Target: Angles of polygons

A geometric figure scaled by factor $k = 3$ has an original side length of 6 cm. What is the corresponding image length?



a. 15 cm

b. 18 cm

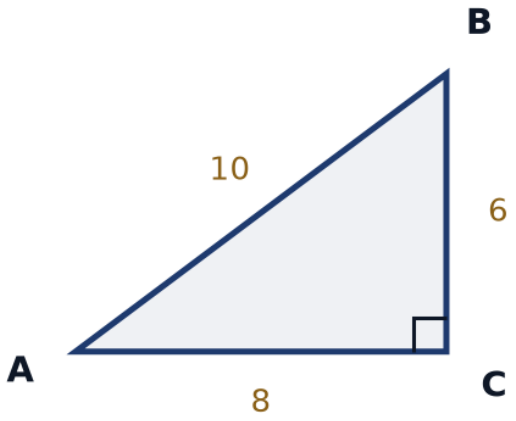
c. 24 cm

d. 54 cm

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23 [Foundational] • Target: Angles of polygons

On the coordinate plane, a line with slope $m = 2$ passes through $(0, 6)$. What is its y-coordinate when $x = 3$?



a. 10

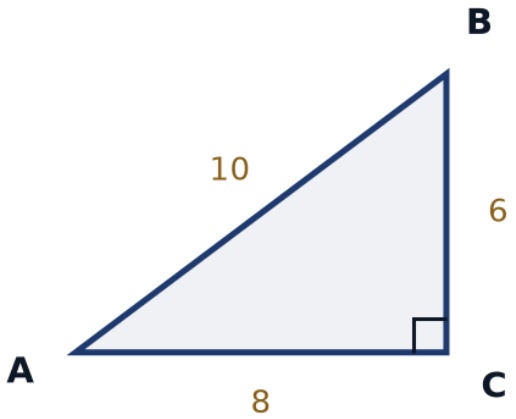
b. 14

c. 12

d. 14

24 [Foundational] • Target: Angles of polygons

In ΔABC studying angles of polygons, $m\angle A = 32^\circ$ and $m\angle B = 66^\circ$. What is $m\angle C$?



a. 82°

b. 92°

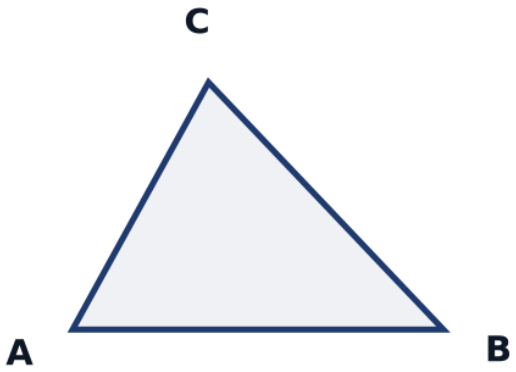
c. 74°

d. 148°

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25 [Foundational] • Target: Properties of parallelograms

In ΔABC studying properties of parallelograms, $m\angle A = 33^\circ$ and $m\angle B = 45^\circ$. What is $m\angle C$?



a. 102°

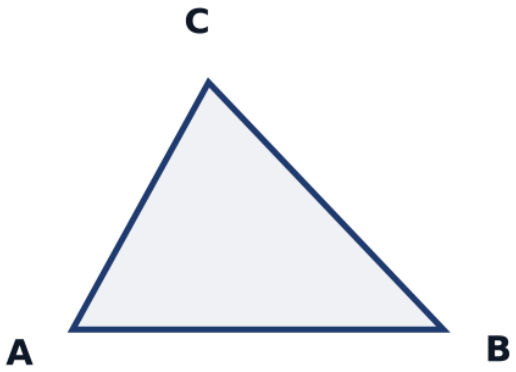
b. 112°

c. 94°

d. 147°

26 [Foundational] • Target: Properties of parallelograms

A geometric figure scaled by factor $k = 3$ has an original side length of 7 cm. What is the corresponding image length?



a. 18 cm

b. 21 cm

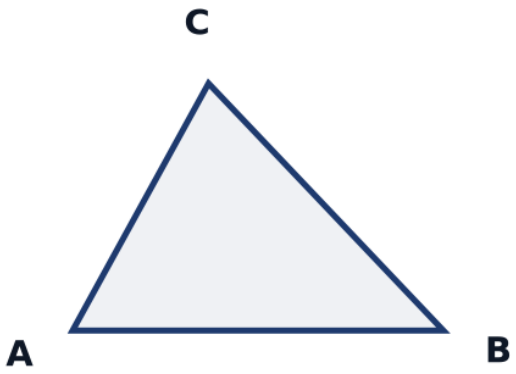
c. 28 cm

d. 63 cm

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27 [Foundational] • Target: Properties of parallelograms

On the coordinate plane, a line with slope $m = 3$ passes through $(0, 7)$. What is its y-coordinate when $x = 3$?



a. 13

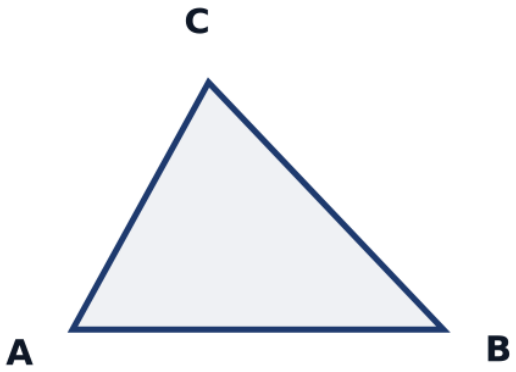
b. 18

c. 16

d. 19

28 [Foundational] • Target: Properties of parallelograms

In ΔABC studying properties of parallelograms, $m\angle A = 33^\circ$ and $m\angle B = 66^\circ$. What is $m\angle C$?



a. 81°

b. 91°

c. 73°

d. 147°

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

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29 [Intermediate] • Target: Angles of polygons

A geometric figure scaled by factor $k = 2$ has an original side length of 6 cm. What is the corresponding image length?

a. 10 cm

b. 12 cm

c. 18 cm

d. 24 cm

30 [Intermediate] • Target: Angles of polygons

In ΔABC studying angles of polygons, $m\angle A = 32^\circ$ and $m\angle B = 52^\circ$. What is $m\angle C$?

a. 96°

b. 106°

c. 88°

d. 148°

31 [Intermediate] • Target: Angles of polygons

In ΔABC studying angles of polygons, $m\angle A = 32^\circ$ and $m\angle B = 59^\circ$. What is $m\angle C$?

a. 89°

b. 99°

c. 81°

d. 148°

32 [Intermediate] • Target: Angles of polygons

A geometric figure scaled by factor $k = 5$ has an original side length of 6 cm. What is the corresponding image length?

a. 25 cm

b. 30 cm

c. 36 cm

d. 150 cm

33 [Intermediate] • Target: Properties of parallelograms

A geometric figure scaled by factor $k = 2$ has an original side length of 7 cm. What is the corresponding image length?

a. 12 cm

b. 14 cm

c. 21 cm

d. 28 cm

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34 [Intermediate] • Target: Properties of parallelograms

In ΔABC studying properties of parallelograms, $m\angle A = 33^\circ$ and $m\angle B = 52^\circ$. What is $m\angle C$?

a. 95°

b. 105°

c. 87°

d. 147°

35 [Intermediate] • Target: Properties of parallelograms

A geometric figure scaled by factor $k = 4$ has an original side length of 7 cm. What is the corresponding image length?

a. 24 cm

b. 28 cm

c. 35 cm

d. 112 cm

36 [Intermediate] • Target: Properties of parallelograms

On the coordinate plane, a line with slope $m = 3$ passes through $(0, 7)$. What is its y-coordinate when $x = 4$?

a. 16

b. 21

c. 19

d. 22

37 [Intermediate] • Target: Properties of rectangles, rhombuses, and squares

On the coordinate plane, a line with slope $m = 4$ passes through $(0, 8)$. What is its y-coordinate when $x = 1$?

a. 8

b. 14

c. 12

d. 16

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

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38 [Challenging] • Target: Properties of rectangles, rhombuses, and squares • Cross-Lesson: 6.01 Polygons & Quadrilaterals

On the coordinate plane, a line with slope $m = 4$ passes through $(0, 8)$. What is its y-coordinate when $x = 2$?

a. 12

b. 18

c. 16

d. 20

39 [Challenging] • Target: Properties of rectangles, rhombuses, and squares • Cross-Lesson: 7.01 Circles

A geometric figure scaled by factor $k = 4$ has an original side length of 8 cm. What is the corresponding image length?

a. 28 cm

b. 32 cm

c. 40 cm

d. 128 cm

40 [Challenging] • Target: Properties of rectangles, rhombuses, and squares • Cross-Lesson: 8.01 Area, Surface Area & Volume

On the coordinate plane, a line with slope $m = 4$ passes through $(0, 8)$. What is its y-coordinate when $x = 4$?

a. 20

b. 26

c. 24

d. 28

41 [Challenging] • Target: Classifying quadrilaterals • Cross-Lesson: 1.01 Foundations & Reasoning

A data set recorded for classifying quadrilaterals consists of the four values $\{12, 16, 20, 26\}$. What is the arithmetic mean?

a. 16.5

b. 17.5

c. 18.5

d. 20.5

42 [Challenging] • Target: Classifying quadrilaterals • Cross-Lesson: 10.01 Coordinate Geometry

For the model $f(x) = 3x + 3$ in classifying quadrilaterals, evaluate $f(4)$.

a. 15

b. 18

c. 12

d. 19

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43 [Challenging] • Target: Classifying quadrilaterals • Cross-Lesson: 11.01 Probability

Find the positive solutions to the quadratic equation $x^2 - 10x + 24 = 0$.

a. $x = 4$ and $x = 7$

b. $x = 4$ and $x = 6$

c. $x = -4$ and $x = -6$

d. $x = 10$ and $x = 24$

44 [Challenging] • Target: Coordinate methods with quadrilaterals • Cross-Lesson: 2.01 Lines & Angles

A data set recorded for coordinate methods with quadrilaterals consists of the four values $\{13, 17, 21, 27\}$. What is the arithmetic mean?

a. 17.5

b. 18.5

c. 19.5

d. 21.5

45 [Challenging] • Target: Coordinate methods with quadrilaterals • Cross-Lesson: 3.01 Triangles & Congruence

For the model $f(x) = 4x + 4$ in coordinate methods with quadrilaterals, evaluate $f(4)$.

a. 20

b. 24

c. 16

d. 24

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